

RAPID PROTOTYPING





INTRODUCTION

- The competition in the world market for manufactured product has intensified tremendously in recent years.
- In order to stay competitive in the high- tech electronics consumer market, companies must continue to offer new products with superior capabilities, higher power efficiency, and smaller form factors at accelerated design schedules.

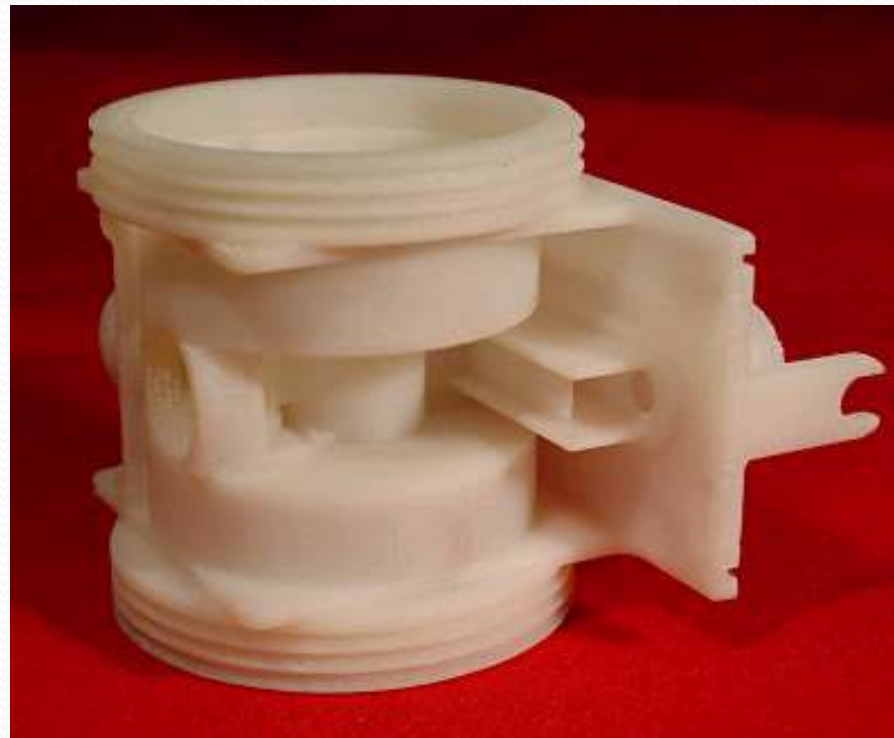


INTRODUCTION

- To bring product to the market quickly, many of the processes involved in the design, test , manufacture and market of the products has been managed ,both in terms of time and material resources.
- They are mainly technology driven , usually involving the computer.

PROTOTYPE

- A prototype is an important and vital part of the product development process.
- It is a model or preliminary version.





ROLES OF THE PROTOTYPES

- Experimentation and learning
- Testing and Proofing
- Communication and interaction
- Synthesis and integration
- Scheduling and markers



PROTOTYPING

- It is the process of realizing the prototypes.
- The intention of having a physical prototype is to realize the conceptualization of a design.
- It is required before the start of the full production of the product.

THREE PHASES OF DEVELOPMENT LEADING TO RPT

- First Phase :Manual Prototyping

The techniques used in making these prototypes tend to be craft based and are extremely labor intensive.

- Second phase :Soft or Virtual Prototyping

It involves using computer aided design (CAD) and computer aided engineering (CAE) software to validate a design

- Third Phase :Rapid Prototyping

It is called Desktop manufacturing Layer manufacturing



THE FATHERS OF RAPID PROTOTYPING

- In the early 1980's **Charles Hall** developed the liquid polymerization based Stereolithography apparatus (SLA)
- Granted the original patent
- Considered to be “the founder” of RP
- In the late 1987, **Carl Deckard** developed the powder based Selective Layer Sintering process (SLS)

RAPID PROTOTYPING

- Group of techniques used to quickly fabricate a scale model of a part or assembly using three dimensional computer aided design data.
- Rapid Prototyping ,also referred to as solid free-form manufacturing, computer automated manufacturing, and layered manufacturing.
- RP has obvious use as a vehicle for visualization.

RAPID PROTOTYPING

Technology which considerably speeds the iterative product development process

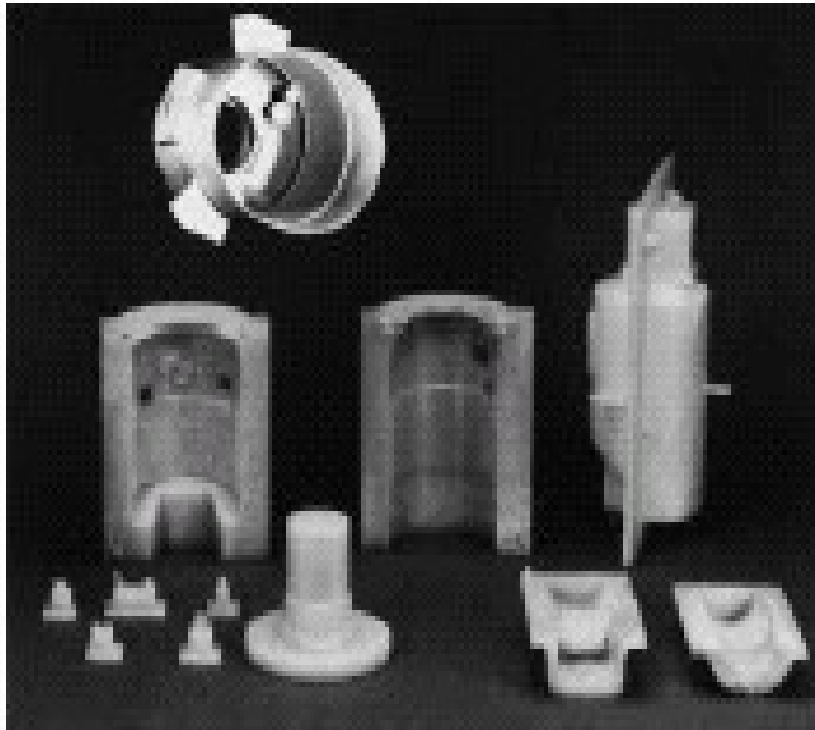


Fig: a) Examples of parts made by rapid prototyping b) cellular phone



RAPID PROTOTYPING

- Rapid Prototyping (RP) techniques are methods that allow designers to produce physical prototypes quickly.
- It consists of various manufacturing processes by which a solid physical model of part is made directly from 3D CAD model data without any special tooling.
- The first commercial rapid prototyping process was brought on the market in 1987.
- Nowadays, more than 30 different processes (not all commercialized) with high accuracy and a large choice of materials exist.
- These processes are classified in different ways: by materials used, by energy used, by lighting of photopolymers, or by typical application range.

WHY RP ?

- To decrease development time
- To decrease costly mistakes
- Increasing product complexity
- More organic , sculptured shapes for functional or aesthetic reasons can be accommodated.
- Designers can optimize part design to meet customer requirements with little manufacturing restrictions.

WHY RP ?

- They can minimize time consuming discussions and evaluations of manufacturing possibilities.
- They can reduce parts count by combining features in one piece part.
- They can minimize material and optimize strength/weight ratio without regard to the cost of machining.
- The manufacturing engineer can minimize design, manufacturing and verification of tooling.

WHY RP

- The manufacturer can reduce labor content of manufacturing.
- To minimize sustaining engineering changes.
- Reduces capital equipment.
- Reducing material waste, waste disposal costs, material transportation costs ,inventory costs for raw stock and finished parts .
- Less effort in scheduling production
- One can reduce inspection rejection rate.



KEY ASPECTS OF RPT

- INPUT
- METHOD
- MATERIAL
- APPLICATIONS

FUNDAMENTAL AUTOMATED PROCESSES

- Subtractive
- Additive
- Formative
 - In subtractive we start with stock and remove material to get the net shape (traditional machining)
 - In additive we start with nothing and add material to get the net shape

Methodology Or Process chain

- CAD solid (surface) model
- '.STL' file
- Slicing the file
- Final build data
- Building (3D printing) the part
- Supports removal
- Post processing

The Rapid Prototyping Technique

- CAD Model Creation:
 - First, the object to be built is modeled using a Computer-Aided Design (CAD) software package.
 - Solid modelers, such as Pro/ENGINEER, tend to represent 3-D objects more accurately than wire-frame modelers such as AutoCAD, and will therefore yield better results.
 - This process is identical for all of the RP build techniques.

The Rapid Prototyping Technique

- **Conversion to STL Format:**

- To establish consistency, the STL (stereolithography, the first RP technique) format has been adopted as the standard of the rapid prototyping industry.
- The second step, therefore, is to convert the CAD file into STL format. This format represents a three-dimensional surface as an assembly of planar triangles
- STL files use planar elements, they cannot represent curved surfaces exactly. Increasing the number of triangles improves the approximation

The Rapid Prototyping Technique

- The parameters used to generate this file will directly influence the quality of the model.
- STL is the standard file type used by all Rapid Prototyping systems.
- Software generates a tessellated object description
- File consists of the X, Y, Z coordinates of the three vertices of each surface triangle, with an index to describe the orientation of the surface normal

The Rapid Prototyping Technique

- **Slice the STL File:**

- In the third step, a pre-processing program prepares the STL file to be built.
- The pre-processing software slices the STL model into a number of layers from 0.01 mm to 0.7 mm thick, depending on the build technique.
- The program may also generate an auxiliary structure to support the model during the build. Supports are useful for delicate features such as overhangs, internal cavities, and thin-walled sections.

The Rapid Prototyping Technique

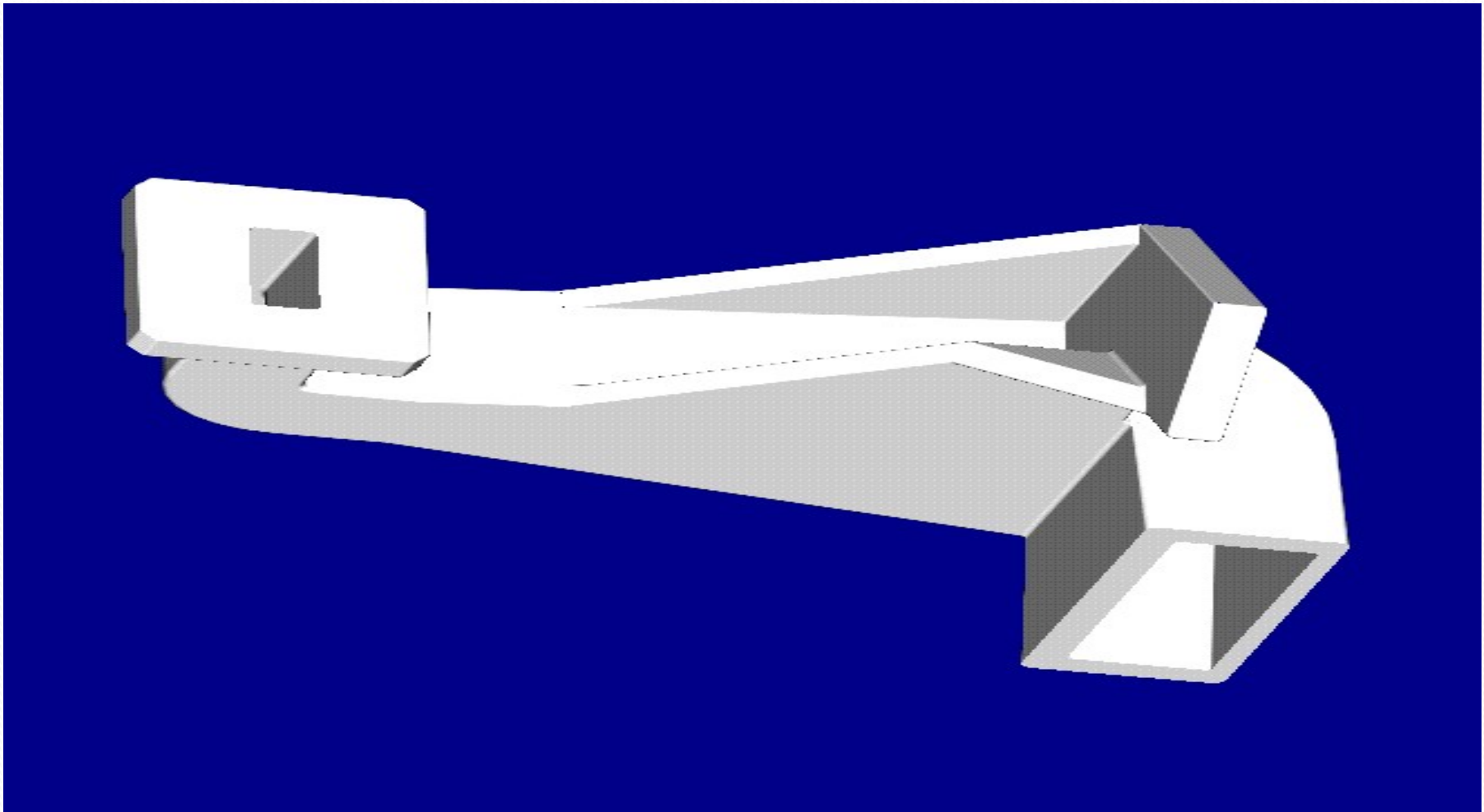
- **Layer by Layer Construction:**
 - The fourth step is the actual construction of the part.
 - RP machines build one layer at a time from polymers, paper, or powdered metal.
 - Most machines are fairly autonomous, needing little human intervention.

The Rapid Prototyping Technique

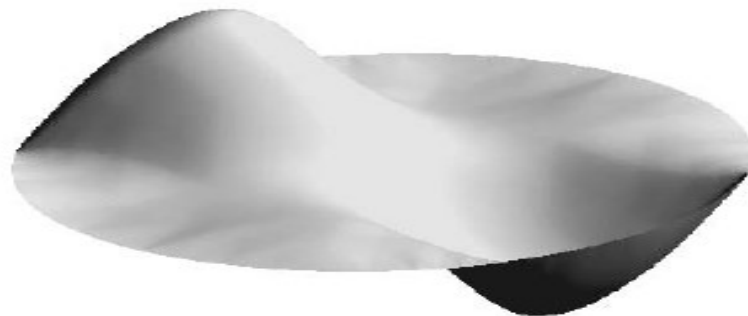
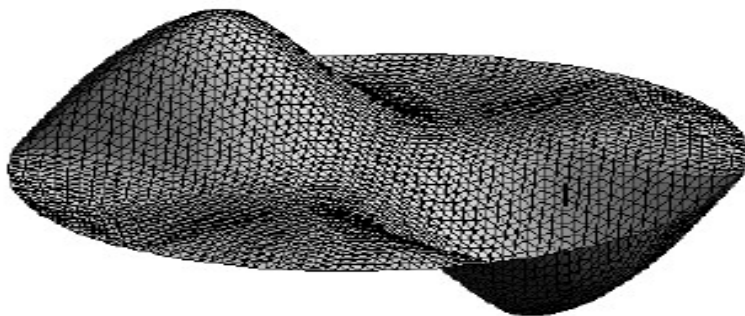
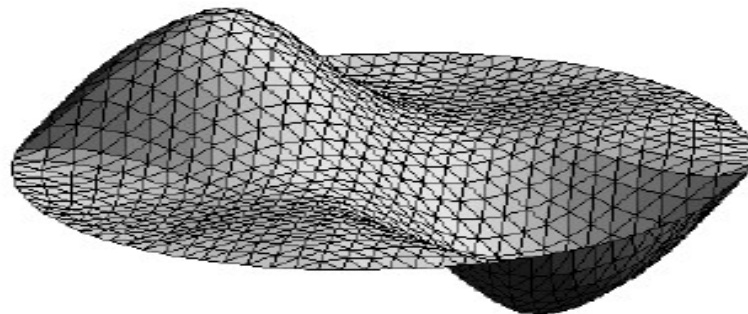
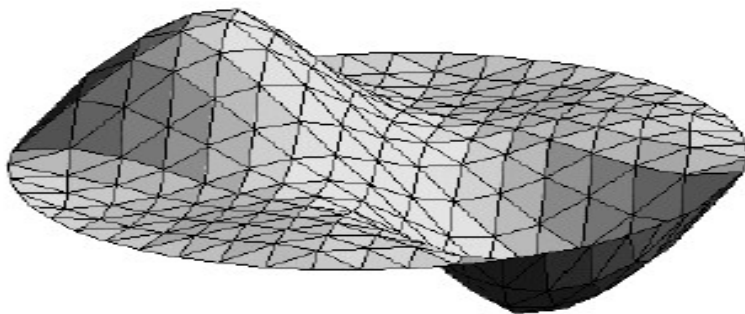
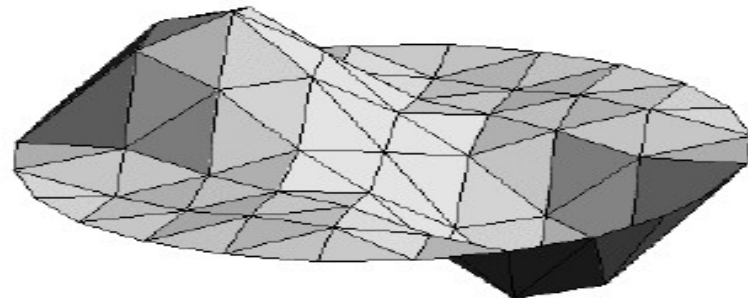
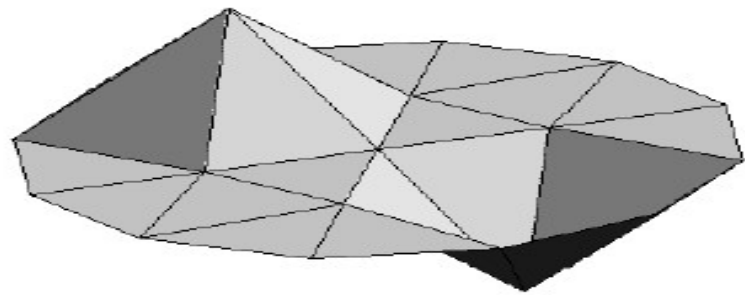
- **Clean and Finish:**

- The final step is post-processing. This involves removing the prototype from the machine and detaching any supports.
- Some photosensitive materials need to be fully cured before use
- Prototypes may also require minor cleaning and surface treatment.
- Sanding, sealing, and/or painting the model will improve its appearance and durability.

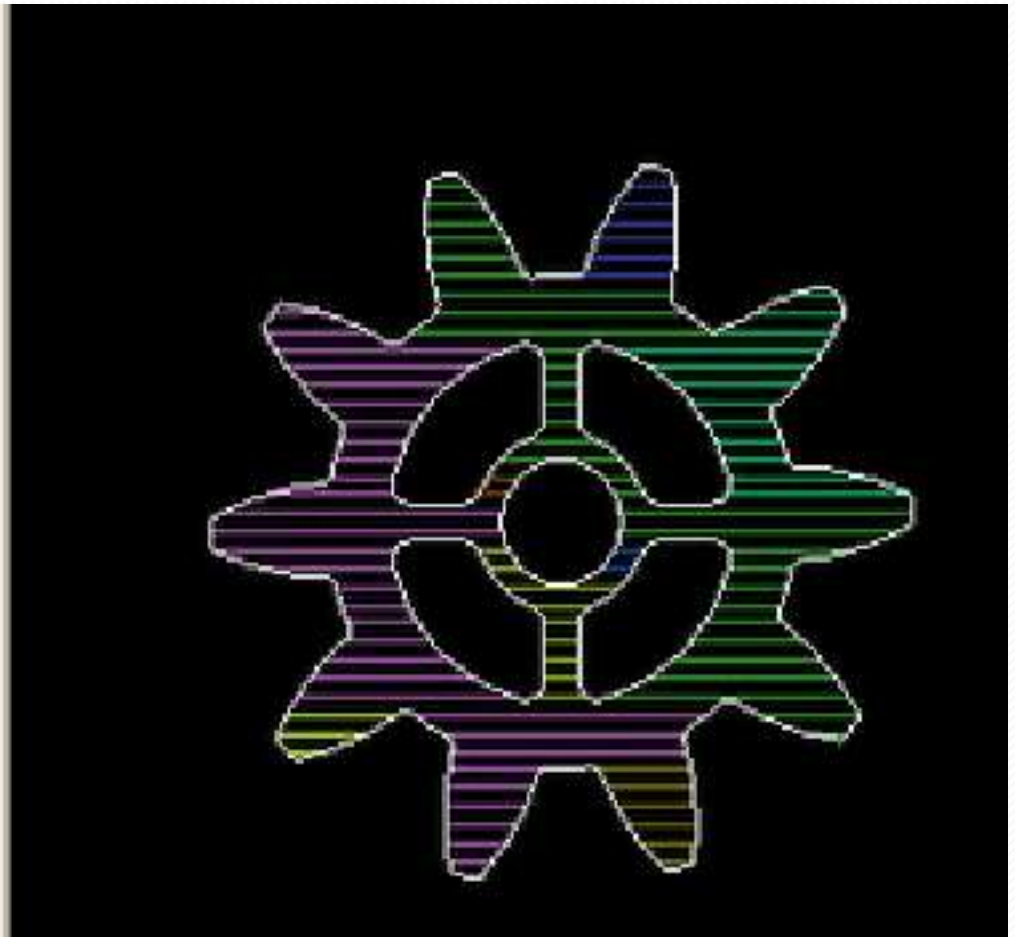
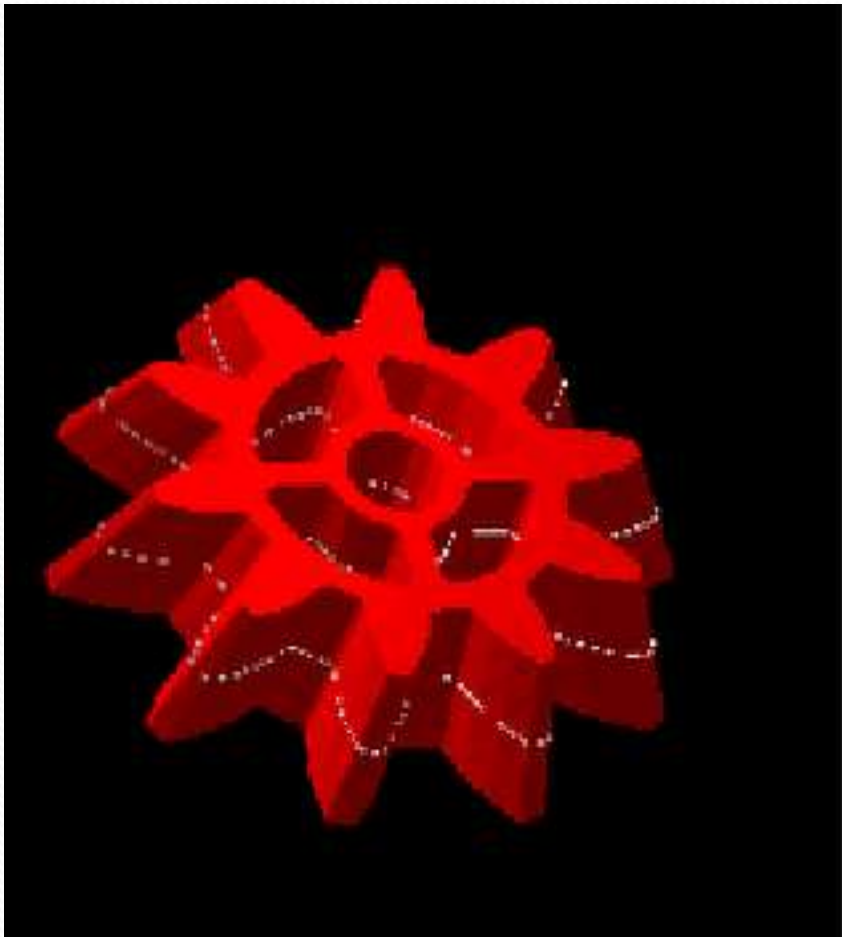
CAD Solid Model



CAD TO STL



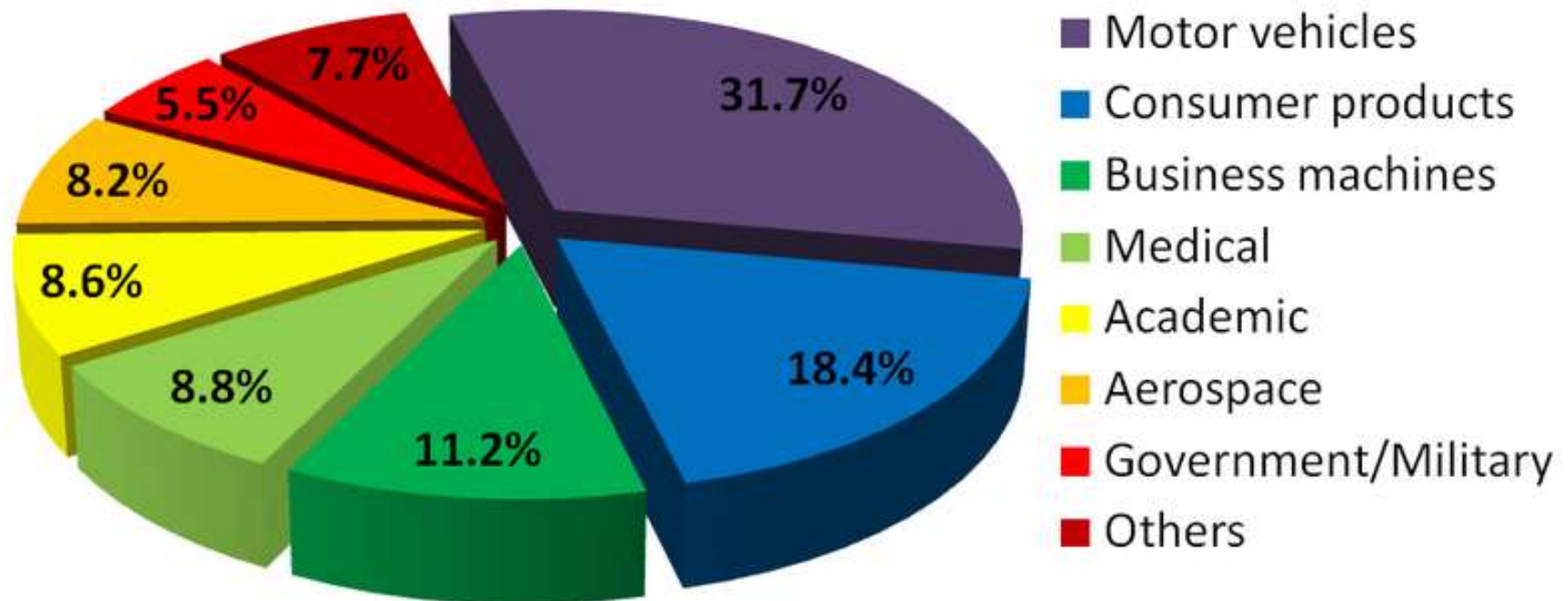
Slicing the model –contour line



RP MODEL



RP USERS



Rapid Prototyping

- Regarding the application level “rapid prototyping”, two sub-levels can be distinguished:
 - “Solid Imaging” and “Concept Modeling” on the one hand and “Functional Prototyping” on the other hand.
- Solid imaging or concept modeling defines a family of parts that are applied to verify a basic concept.
- The parts resemble a three dimensional picture or a statue. In most cases, they cannot be loaded.
- They are used just to get a spatial impression in order to judge the general appearance and the proportions.
- Because of this the parts are also called “show and tell models”

Functional Prototyping

- It is applied to allow checking and verifying one or more isolated functions of the later product or to make the production decision, even though the model cannot be used as a final part.
- As can be seen in figure below, the adjustable air outlet grill for the climate adjustment nozzle of a passenger car can be used to verify the air distribution in a very early stage of product development. It was made in one piece using laser stereolithography.

Functional prototyping:
adjustable air outlet grill for a passenger car;
laser stereolithography

(Source: 3D Systems)



Functional Prototyping

Test form, fit, and function with models built in tough ABS plastic for functional prototyping



RAPID MANUFACTURING & RAPID TOOLING



What is Rapid Manufacturing?

- We try to take the customer feedback or requirements converted it into a digital form; and that digital form is given directly to the machine the parts are produced and directly moved towards an application.
- So, we are not going to go into a field called as prototyping, we make a prototype for that prototype then we make tools and from that tools we try to go for the final product.
- So, Rapid Manufacturing is more towards mass customization where cost is not to be compromised, but things have to be produced to the customers satisfaction with the highest integrity, that is Rapid Manufacturing.

Why Rapid Manufacturing?

- Industrial 3-D printing is at a tipping point, about to go mainstream in a big way.
- Most executives and many engineers don't realize it, but this technology has moved well beyond prototyping, rapid tooling, trinkets, and toys.
- "Rapid manufacturing" is creating durable and safe products for sale to real customers in moderate to large quantities.

Why Rapid Manufacturing?

Shrinking product manufacturing times:

- The assembly line process revolutionized production and dropped the assembly time for a single vehicle from 12 hours to a few minutes.
- Ford Freeform Fabrication Technology, or F3T, will lower costs and speed delivery times for prototype stamping molds – within three business days versus two against six months for prototypes made using conventional methods.
- Through Ford's "virtual factory," the company can improve quality and cut costs in real-world manufacturing facilities by creating and analyzing computer simulations of the complete vehicle production process



Rapid Manufacturing

- The application level “Rapid Manufacturing” summarizes all processes that deliver final products or final parts that need to be assembled to become a product.
- An AM part is called a product or final part, if it shows all characteristics and functions that are allocated to it during the product development process.
- If the resulting part is a positive, the process is called “Direct Manufacturing”, if it is a negative, which means a die, a mold or a gauge, it is named “Direct Tooling”.
- Direct Manufacturing leads to final parts that directly come from the AM process (as mentioned in the illustration)

Rapid Manufacturing

- Today, a broad variety of materials from all material classes (plastics, metals, and ceramics) are available to be processed directly using an AM process.
- Here, it is not important that the available materials show exactly the same physical properties as the materials used within traditional fabrication processes.
- However, it must be assured that the properties on which the engineering design was based can be realized with the chosen AM process and material.

Rapid Manufacturing

- The data was obtained from the patient by a dental imprint and then digitized.
- With the use of a professional dental software (3shape) the dental bridge was designed and directly manufactured using SLM.
- After finishing and geometric testing, the bridge was ready to be placed in the patient's mouth.
- Regarding traditional processes, the directly manufactured bridge was made quicker, with a perfect fit, and at comparable cost.



Rapid Tooling

- Rapid Tooling involves all AM procedures that lead to final parts used as cores, cavities, or inserts for tools, dies and molds.
- Two sub-levels must be distinguished:
 - direct tooling
 - prototype tooling.
- *Direct Tooling* is technically equivalent to *Direct Manufacturing* but leads to tool inserts, dies and molds in series quality.
- Although tooling is based simply on the inversion of the product data set (positive to negative) there are reasons to attribute it with a separate application sub-level

Rapid Tooling

- In addition, for data inversion, a tool construction is needed, including scaling to compensate shrinking, parting line definitions, draft angles, ejectors, sliders, and so on.
- Tooling mostly requires a metal process and machines that are designed to run it.
- It is important to understand that “Direct Tooling” does not mean that the entire tool is made, in fact only tool components, such as cavities or sliders, are generated.
- The entire tool is made using these cavities and standard components or inserts within a traditional tool making process.

Rapid Tooling

Direct tooling. Mold insert with conformal cooling channels (blue) and pneumatic ejectors (white).

Laser sintering/laser melting (laser cusing);
tool steel

(Source: Concept Laser)



Prototype Tooling

- A mold in series quality often is too time and money consuming for small series manufacturing.
- If just a few parts are needed or details are changed frequently, a temporary mold made from substitute material is typically sufficient.
- This kind of mold shows the quality of functional prototypes but meets, at least partially, the direct tooling application level.
- The corresponding application level is some kind of an intermediate level between rapid prototyping and rapid manufacturing.
- This sub-level is called “Prototype Tooling”

Prototype Tooling

- Some call it “Bridge Tooling”, although this name is also used for secondary rapid prototyping processes.
- A prototype tool made from polyamide can be seen as an example in figure below.
- It is used to fabricate a small series of a new design for a grip sole for rubber boots.

Prototype tooling; rubber boot sole mold;
laser sintering, polyamide (PA)
(Source: EOS GmbH)



Prototype Tooling

- Summing up the different kinds of tooling, one can see that “Rapid Tooling” does not represent an autonomous application level.
- Rapid tooling integrates all AM applications that can manufacture dies and molds or corresponding inserts.
- The soles are necessary to make the entire boot by applying it to a prefabricated bootleg without making a series-like metal tool in advance.

Direct Processes

- All AM processes are called “direct processes” in order to indicate that the digital process model is directly converted into a physical object, called the part, by means of a generative machine.
- In contrast to this, some procedures are called “indirect processes” or also “indirect rapid prototyping processes”. They do not apply the principle of layer manufacturing and consequently they are not AM processes.
- Actually indirect processes are copying techniques mostly based on silicon rubber casting such RTV.
- Since AM parts are used as masters, the term “Indirect Rapid Prototyping Processes”, was established because it sounds more innovative.

Application Levels – Indirect Processes

The AM process directly delivers a geometrically exact and scaled physical facsimile of the virtual data set. But this process also comes with disadvantages (at least with regard to most of today's AM processes).

- AM processes work with process- and consequently with machine-depending materials and restrictions in terms of color, transparency, and flexibility.
- Show almost no cost reduction with an increasing production volume.
- Are rather expensive when used to make many copies and especially for series applications.

Application Levels – Indirect Processes

- To overcome these problems, AM parts can be regarded as master models and then used for subsequent copying or reproduction processes.
- The principle behind this is often called 'the splitting of capabilities'.
- The geometrical exact part is quickly obtained from the AM process, while the desired quantity, and properties such as color and so on, come from a subsequent copying process.
- A copying- or follow-up process is not a layer-based process and therefore it is not an AM process. It is called an "Indirect Process".

Application Levels – Indirect Processes

- Because of marketing reasons and in order to indicate the manufacturing speed, some call it “Indirect Rapid Prototyping Process” as well.
- For the same reason, in literature sometimes the term “Secondary Rapid Prototyping Process” is used.

Indirect Prototyping

- Indirect prototyping is applied to improve the AM part's properties in order to fulfill the applicator's requirements, if the AM part is not capable to do so.
 - for example:
 - a flexible part is needed but due to material restrictions it cannot be built directly by an AM process.
- Then a geometrical exact but rigid AM part is built (and maybe scaled according to a possibly shrinkage during casting) and used as a master model for a subsequent or follow-up casting process.

Indirect Prototyping

- Many different “secondary processes” can be used.
- The most prominent one is the so called “Room Temperature Vulcanization”, RTV, also known as “Vacuum Casting” or “Silicon Rubber Molding”.
- Like silicon rubber molding, most of the secondary processes are completely or partially manual processes with long cycle times and therefore only useful for small series or one-of-a-kind production.
 - For instance, the plug system requires plug housings of different colors and transparency.
 - Based on a two-parts AM master of the housing, a silicon rubber mold is to be made.

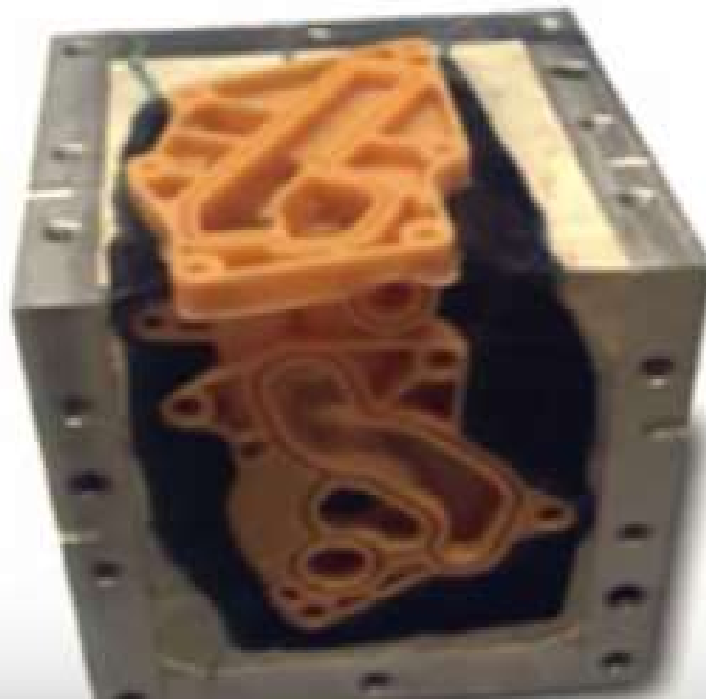
Indirect Tooling

- Indirect tooling is based on the same copying procedures as all indirect processes.
- It is not the goal to obtain a final part, but a tool that provides the basis for a small or medium size batch production of final (or series) parts or products.
- In contrast to series tools made from tool steel, it can be made quickly and inexpensively.
- Like indirect prototyping, indirect tooling uses AM masters, thus avoiding milling, grinding, and EDM-processes.

Indirect Tooling

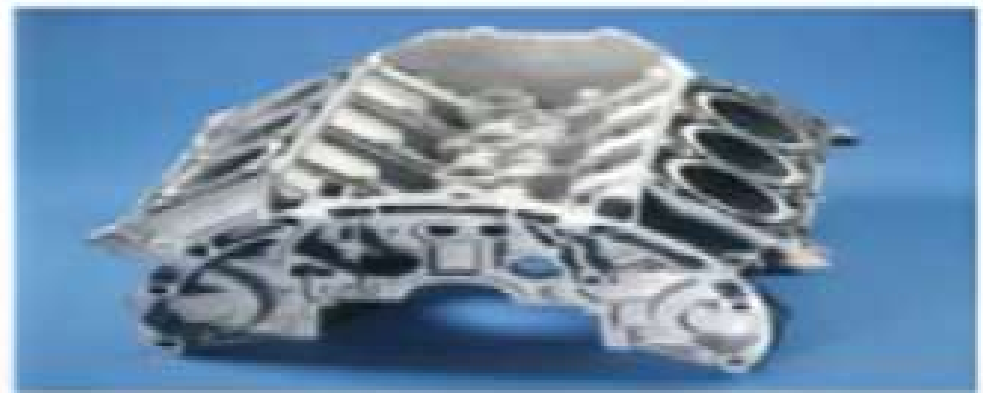
Indirect tooling. PUR mold obtained from an AM master, partly open.
Separated mold half with cast wax pattern for lost wax casting.
PUR cavity (black) obtained from AM master.
Back up system by aluminum walls

(Source: BeNe)



Indirect Manufacturing

- Indirect manufacturing is based on AM masters as well.
- The goal is to obtain final (or series) parts with properties equal to traditionally manufactured (non-AM) products.
- Consequently, indirect manufacturing belongs to the application level “Manufacturing” (Fig. in next slide).
- As an example for indirect manufacturing, Fig. below shows a six-cylinder combustion engine housing.



Indirect Manufacturing

- It was produced as a one-of-a-kind part based on an AM master made from polystyrene by laser sintering.
- The scaled master was transformed into an aluminum part by evaporative pattern casting (also called full-mold casting), which is a process closely related to the lost-wax-casting process.
- As a result, a series of identical engine housings is obtained. It can be used to optimize and verify the engine design, including fired test runs long before series molds are available, but also as a small series product, e.g., for racing.



DEFINITIONS

- **Additive Manufacturing (AM)** is a layer-based automated fabrication process for making scaled 3-dimensional physical objects directly from 3D-CAD data without using part-dependent tools.
- **Rapid Prototyping** is one of the application levels of AM where two sub-levels can be distinguished:
 - **Solid imaging** (or concept modeling) defines a family of parts that are applied to verify a basic concept. The parts resemble a three dimensional picture or a statue.
 - **Functional prototyping** is applied to allow checking and verifying one or more isolated functions of the later product or to make the production decision, even though the model cannot be used as a final part.

Recollecting the definitions

- **Rapid Manufacturing** at the application level, *summarizes all processes* that deliver final products or final parts that need to be assembled to become a product.
 - If the resulting part is a positive, the process is called **Direct Manufacturing**.
 - If it is a negative, which means a die, a mold or a gauge, it is named **Direct Tooling**.
- **Rapid tooling** involves all AM procedures that lead to final parts used as cores, cavities, or inserts for tools, dies and molds.
- **Prototype tooling** is the corresponding application level, some kind of an intermediate level between rapid prototyping and rapid manufacturing.

- **Indirect prototyping** is applied to improve the AM part's properties in order to fulfill the applicator's requirements, if the AM part is not capable to do so.
- **Indirect tooling** is based on the same copying procedures as all the indirect processes. It is not the goal to obtain a final part, but a tool that provides the basis for a small or medium size batch production of final (or series) parts or products.
- **Indirect manufacturing** is based on AM masters as well. The goal is to obtain final (or series) parts with properties equal to traditionally manufactured (non-AM) products.